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Proceedings of the Combustion Institute 33 (2011) 609–616
Proceedings of the Combustion Institute
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Soot formation characteristics of gasoline surrogate fuels in counterflow diffusion flames

B.C. Choi, S.K. Choi, S.H. Chung *

Clean Combustion Research Center, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

Available online 23 August 2010

Abstract

The characteristics of polycyclic aromatic hydrocarbon (PAH) and soot for gasoline surrogate fuels have been investigated in counterflow diffusion flames by adopting laser-induced fluorescence (LIF) and laser-induced incandescence (LII) techniques for both soot formation and soot formation/oxidation flames. Tested fuels were three binary mixtures from the primary reference fuels of *n*-heptane, *iso*-octane, and toluene. The result showed that PAH and soot maintained near zero level for all mixtures of *n*-heptane/*iso*-octane case under present experimental conditions. For *n*-heptane/toluene and *iso*-octane/toluene mixtures, PAH initially increased and then decreased with the toluene ratio, exhibiting a synergistic effect. The soot formation increased monotonically with the toluene ratio, however the effect of toluene on soot formation was minimal for relatively small toluene ratios. These results implied that even though toluene had a dominant role in soot and PAH formations, small amount of toluene had a minimal effect on soot formation. Numerical simulations have also been conducted by adopting recently proposed two kinetic mechanisms. The synergistic behavior of aromatic rings was predicted similar to the experimental PAH measurement, however, the degree of the synergistic effect was over-predicted for the soot formation flame, indicating the need for refinements in the kinetic mechanisms.
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Keywords: Soot; Polycyclic aromatic hydrocarbon (PAH); Laser-induced incandescence; Laser-induced fluorescence; Gasoline surrogate fuel

1. Introduction

Gasoline fuel occupies a significant portion of transportation fuels. Because of recent issues on energy and environmental concerns, gasoline fuel formulation becomes an interesting topic. There are numerous compositions in gasoline fuels and engine emission is appreciably influenced by fuel composition [1]. Although there were many studies on the effect of fuel composition on engine emissions, details of the effect on fundamental flame characteristics are rather limited. Gasoline fuel is a compound of various hydrocarbon species. Among them, representative components can be regarded as *n*-heptane (n-C_7H_{16}), *iso*-octane (iso-C_8H_{18}), and toluene (C_7H_8) as the surrogate fuels. Studies on these surrogate fuels have been frequently focused on ignition delay time, autoignition, and extinction characteristics [2–7].

Recently, direct-injection spark-ignition (DISI) engines have been extensively developed for fuel efficiency and emission control. However, direct injection could promote the formation of particulate matter (PM). Although the amount of PM

* Corresponding author.
E-mail address: sukho.chung@kaust.edu.sa (S.H. Chung).

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doi:10.1016/j.proci.2010.06.067

emission in DISI engines is expected to be much less than that in diesel engines, it could pose problems in the future because of the regulation trend of PM by numbers and the general trend to avoid the installation of particulate filter. Therefore, the study on the effect of fuel composition on soot formation characteristics for gasoline fuels can be one of the important issues.

Sooting characteristics in counterflow diffusion flames are markedly different depending on the relative position of flame to the stagnation plane [8]. When the flame is located on the fuel side (Fig. 1 inset), since soot zone is located on the fuel side of the flame, once soot particles are formed, they will migrate toward the flame and subsequently oxidized by the diffusion of oxidizing species such as OH and O_2. This type of flame is categorized as the soot formation/oxidation (SFO) flame. While when the flame is located on the oxidizer side, soot particles will be convected away from the flame such that soot oxidation cannot take place and hence classified as the soot formation (SF) flame.

In the present study, the characteristics of soot and polycyclic aromatic hydrocarbons (PAHs) are examined for the surrogate fuels of the binary combinations of *n*-heptane, *iso*-octane, and toluene for both SF and SFO flame conditions. It has been reported that mixture fuels could exhibit a synergistic effect on PAH or soot formation, which is the case when a mixture fuel exhibits enhanced formation as compared to respective pure fuels [9–11]. The occurrence of synergistic effect is explored along with numerical simulations to test the predictability of PAH formation.

2. Experiment

The apparatus consisted of a counterflow burner and flow controllers, a liquid fuel vaporization system, and a visualization setup. The burner had diverging, settling, and converging sections with the area contraction ratio of 36 to maintain near uniform flow at the exit [12]. The diameters of two opposed nozzles were 10 mm with 8 mm separation. Liquid fuels with commercially-pure grade were used for *n*-heptane (99%), *iso*-octane (99.8%) and toluene (99.8%), which were controlled by a syringe (SGE, 50MDR-LL-GT) and pump (KDS100). Gas flow rates were metered by mass flow controllers calibrated by a wet-test gas meter. Nitrogen was used as shield gas and supplied through concentric slits surrounding the nozzles to prevent the influence of ambient air.

To vaporize liquid fuels, the fuel supply line was heated by strip heaters (225 W) to keep the temperature at the bottom of the fuel burner at 180 ± 3 °C. Note that the boiling temperatures of *n*-heptane, *iso*-octane and toluene are 98.5, 99.22, and 110.63 °C, respectively [13]. To prevent condensation of vaporized fuels, the fuel nozzle was heated to keep the temperature at 150 ± 3 °C at the nozzle exit by strip heaters (185 W). The oxidizer was maintained at ambient temperature.

Laser-induced incandescence (LII) and laser-induced fluorescence (LIF) techniques were adopted to measure the relative concentrations of soot and PAH. In the LII measurement, the second harmonic (532 nm) of Nd:YAG laser (Spectra-Physics, LAB170) was used to form a sheet beam with a cylindrical lens (f12.7 mm) and convex lens (f500 mm). The LII signals were obtained at saturation condition above a threshold level at 2000 mJ/cm^2 laser fluence for the signal to be independent of laser fluence [14,15]. To minimize noises in soot scattering, a band pass filter with 400 nm (FWHM 40 nm) was used [10,11] to detect LII signal through an intensified charge-coupled device (ICCD) camera (Princeton instrument, ICCD MAX) with 100 ns gate duration after laser shot without delay. In the PAH LIF experiment, the excitation wavelength was 283 nm by using a Nd:YAG laser (Continuum, PL8000) and dye laser (Continuum, ND6000) with 250 mJ/cm^2 laser fluence. Images were captured by the ICCD camera with 30 ns gate duration after laser shot [10,11]. Band pass filters of 330 (FWHM 10 nm), 400, 450, and 500 nm (FWHM 40 nm) were used to differentiate relative size groups of PAHs [10,11]. It has been qualitatively shown previously that as the detection wavelength becomes longer, the LIF signal represents larger ring aromatics [10,16].

Depending on fuel and oxidizer dilutions with nitrogen, the position of diffusion flame in a counterflow can be controlled. The criterion for the flame to be located on the fuel side is $Y_{O,0}/\sqrt{Le_O} > \sigma Y_{F,0}/\sqrt{Le_F}$ [17], where Y is the mass fraction, Le is the Lewis number, σ is the stoichiometric oxidizer to fuel mass ratio, and the subscripts F, O, and 0 indicate fuel, oxygen, and initial condition, respectively.

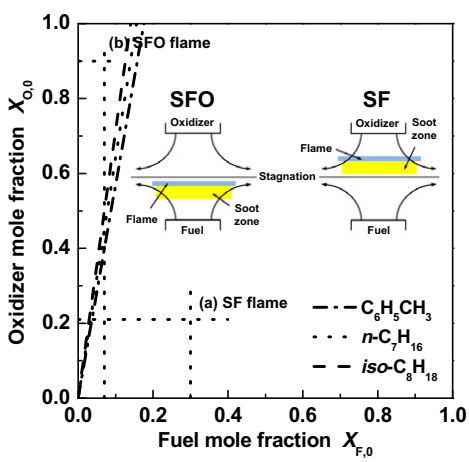

Fig. 1. Experimental conditions for SF (a) and SFO (b) flames.

Using this relation, the condition of a counterflow flame can be determined on a $X_{F,0}$–$X_{O,0}$ plane as demonstrated in Fig. 1, where X indicates the mole fraction. The left (right) side of the boundary is for SFO (SF) flame condition. Note that the boundaries are very similar for the tested fuels. The experimental conditions are selected at $(X_{F,0}, X_{O,0}) = (0.30, 0.21)$ and $(0.07, 0.90)$ for SF (a) and SFO (b) flames, respectively. To maintain the flame stable, the fuel flow velocity $U_{F,0}$ is specified at 28 cm/s and the oxidizer flow velocity $U_{O,0}$ at 21 cm/s. The liquid fuel was vaporized and diluted with nitrogen and the oxidizer consists of oxygen and nitrogen.

Among the three surrogate fuels, the binary mixtures of *n*-heptane/*iso*-octane, *n*-heptane/toluene, and *iso*-octane/toluene have been tested. The flames of *n*-heptane, *iso*-octane, and their mixtures are blue in color and non-sooting based on the LII signals. Therefore, in the following, the flames of *n*-heptane/toluene and *iso*-octane/toluene are focused.

3. Results and discussion

Figure 2 exhibits direct images with false-color, together with LII and LIF (detection at 400 nm) images, for *iso*-octane/toluene mixtures by varying the toluene ratio R_T, which is defined as the liquid volume ratio of toluene in the binary mixture. The LII and LIF images were the realization of 100 frames. Both SF and SFO flames are exhibited. For pure *iso*-octane ($R_T = 0$), the flame has blue color and the LII signal is negligibly small. As the toluene ratio increases, a yellow luminous flame appears by the black body radiation from soot. The yellow flame region expands and the corresponding LII intensity becomes stronger. Both SF and SFO flames have similar tendency. One interesting feature is the LIF images exhibiting double-layer behavior for SFO flames, especially for $R_T = 1.0$. The flames of *n*-heptane/toluene mixtures demonstrate similar trends.

3.1. Soot formation flames

The centerline profiles of LII and LIF (400 nm) signals are exhibited in Fig. 3 for *iso*-octane/toluene flames at several R_T for the SF flame conditions. The data are normalized with the maximum values at $R_T = 1.0$. The vertical dotted line indicates blue flame position near 4.15 mm from the fuel nozzle, which is insensitive to R_T. When the toluene ratio is small up to say 0.4, the variation in LII signal is minimal. For $R_T > 0.4$, the maximum LII signal increases rapidly with R_T.

The maximum PAH LIF locations are on the fuel side of the LII signals, in accordance with soot formation pathway from PAHs to soot. As R_T increases, the maximum PAH signal increases rather rapidly as compared to the LII signals. One interesting feature is that the maximum LIF signal starts to decrease at high toluene ratios, e.g., the maximum for $R_T = 0.6$ and 0.8 are higher than that for 1.0, although the overall profiles and the position of maximum LIF signal are similar. This exhibits a synergistic effect on PAH signals by the toluene ratio. Although not shown, this

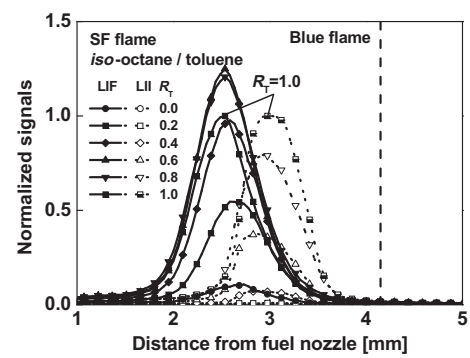

Fig. 3. Profiles of normalized LII and LIF signals for *iso*-octane/toluene mixture flames at several toluene ratios for SF flames.

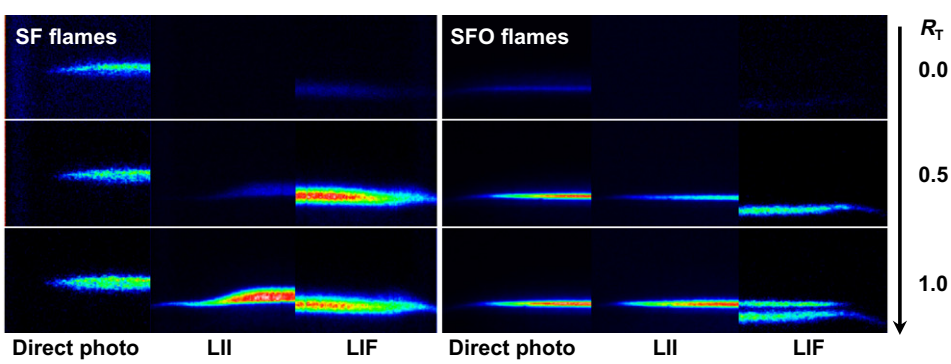

Fig. 2. Direct photos and LII and LIF images with toluene ratio in SF and SFO flames of *iso*-octane/toluene mixtures.

synergistic effect can also be observed for all the LIF detection wave lengths.

The LII signals of the *n*-heptane/toluene mixture fuels show similar behavior, although the signal increases somewhat slower than that for the *iso*-octane/toluene case for small toluene ratios. The LIF signals at 450 and 500 nm show the synergistic effect, while for lower wavelength especially at 330 nm, the effect disappears.

The normalized maximum LII and LIF (400 nm) signals with $R_T = 1$ data are shown in Fig. 4. The LII signals increase with the toluene ratio monotonically, while their behaviors are far from linear. For both *n*-heptane and *iso*-octane cases, the maximum LII signals are minimal up to $R_T \approx 0.4$, and then the signal starts to increase reasonably linearly. With regard to soot formation, this is an interesting result. As mentioned previously, pure fuels of *n*-heptane and *iso*-octane are non-sooting, and the flame with pure toluene is highly sooting. The present results indicate that there exists a tolerance in terms of toluene mixing to either *n*-heptane or *iso*-octane, below which the effect of toluene on soot formation can be regarded as negligible.

The maximum LIF signals exhibit non-monotonic behavior. For small R_T, the maximum signal increases reasonably linearly with the toluene ratio. As R_T further increases, the maximum signals reach a peak value at $R_T \approx 0.7$ and then decrease. Thus, the synergistic behavior in the maximum of LIF signal is clearly exhibited, meaning that the maximum value at certain range of R_T is even larger than the maximum values at $R_T = 0$ and 1.0. This synergistic effect is relatively weaker for the *n*-heptane/toluene mixtures as compared to the *iso*-octane/toluene mixtures.

In the previous studies on soot formation in a counterflow [9–11], the fuel structure effects on synergistic behavior have been observed, for example, for the mixture fuels of ethylene and propane. This has been explained based on the incipient benzene ring formation through propagyl recombination and subsequent hydrogen-abstraction-C_2H_2-addtion (HACA) mechanisms [18–20], that is, based on the aromatic ring formation and the growth of PAH and soot by C_2H_2. The role of methyl radicals on the synergistic effect has been emphasized. The present mixture fuel of *n*-heptane or *iso*-octane and toluene has similar features in that the fuel is the combination of non-aromatic and aromatic fuels.

To elucidate the synergistic behavior of PAHs, numerical simulations have been performed and the formation of aromatic rings is compared with the experimental PAH data. The simulation is performed using OPPDIF code [21] adopting two reaction mechanisms of Blanquart et al. [22] (MEC1) and Marchal et al. [23] (MEC2). These mechanisms have been developed for the modeling of the formation of soot precursors for surrogate fuels, and contain up to four ring aromatics. MEC1 contains 148 species and 928 reactions, and MEC2 154 species and 850 reactions. Thermodynamic and transport properties are calculated by CHEMKIN [24] and TRANSPORT packages [25], respectively.

Figure 5 shows the normalized maximum mole fractions of benzene (A1), naphthalene (A2), phenanthrene (A3), and pyrene (A4) with toluene ratio for the *iso*-octane/toluene mixture. For the benzene ring, both mechanisms exhibit synergistic effect with the peak near $R_T = 0.8$, even though the degree of the synergistic effect is different. For higher poly-aromatic rings, the two mechanisms show different trend. In MEC1, the synergistic behavior disappears for A2, A3 and A4, and the normalized maximum mole fractions increase monotonically with R_T. In MEC2, A2 formation shows complicated trend with R_T, while for A3 and A4, the synergistic effect has been clearly exhibited.

The PAH LIF data at 400 nm has been known to reasonably represent pyrene (A4) [26]. The

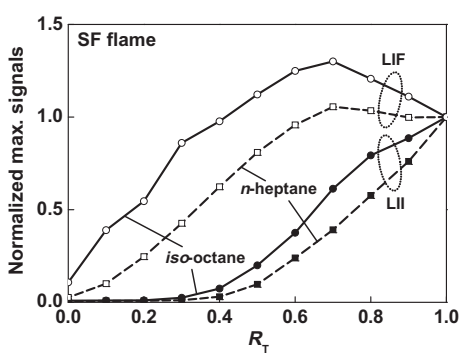

Fig. 4. Normalized maximum LII and LIF signals as a function of toluene ratio in *n*-heptane/toluene and *iso*-octane/toluene mixture flames in SF condition.

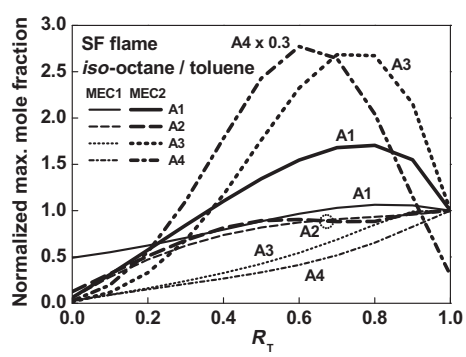

Fig. 5. Normalized maximum mole fractions of A1, A2, A3 and A4 calculated by MEC1 [22] and MEC2 [23] as a function of toluene ratio for *iso*-octane/toluene mixture flames in SF condition.

experiment showed the peak value in the normalized maximum signal of 1.3 at $R_T = 0.7$ for *iso*-octane/toluene mixture. The synergistic behavior with MEC2 is highly over-predicted for A4 in that the peak mole fraction of A4 is about 9(6) times of that for the pure toluene for *iso*-octane(*n*-heptane)/toluene mixture at R_T about 0.6(0.5), although the case of *n*-heptane/toluene mixture is not shown. This over-prediction may be partly attributed to the fact that the consumption of PAHs by soot formation has not been accounted for in the simulation. However, considering that the soot formation is typically in the order of 1% from the total carbon supply in the fuel stream [8,9], the prediction is expected to be an over-estimation. This point will be further discussed in the next section for soot formation/oxidation flames.

Both reaction mechanisms predicted the synergistic behavior in the benzene ring formation. The dominant reaction which produces benzene ring is $C_6H_5CH_3 + H \rightarrow A1 + CH_3$ in both mechanisms. In this regard, Fig. 6 shows the profiles of the mole fractions of benzene, toluene, and H radical at various R_T for the SF flames of *iso*-octane/toluene mixture based on MEC2. The mole fraction of H radical decreases as R_T increases. This can offset the effect of increasing toluene with R_T, leading to the synergistic effect in benzene ring formation such that the mole fraction of benzene for $R_T = 0.8$ is higher than those of 0 and 1.0.

To elucidate the decreasing trend of H radical, the reaction path analysis was performed by ChemPathTool [27] based on Cantera software [28]. Figure 7 shows the reaction path diagram at the position of maximum C_6H_6 for $R_T = 0.8$ in the SF flame of *iso*-octane/toluene mixture. As mentioned previously, benzene ring (A1) is dominantly formed by the reaction of toluene with H radical. When focusing on the H radical, it is produced from *i*-C_3H_7 and *t*-C_4H_9, which originate from *iso*-octane. Therefore, the decrease in *iso*-octane with increasing R_T weakens the formation of H radical even though the concentration of toluene increases. This competing effect results in the synergistic effect in benzene formation.

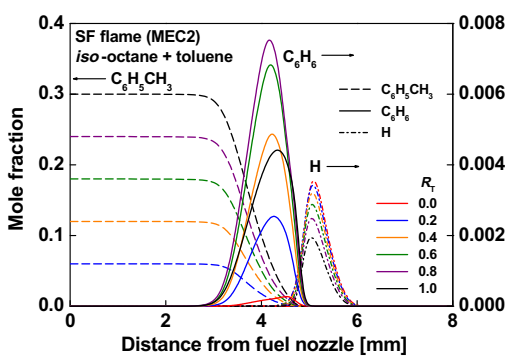

Fig. 6. Profiles of mole fractions of $C_6H_5CH_3$, C_6H_6, and H calculated by MEC2 [23] for various toluene ratios for *iso*-octane/toluene mixture flames in SF condition.

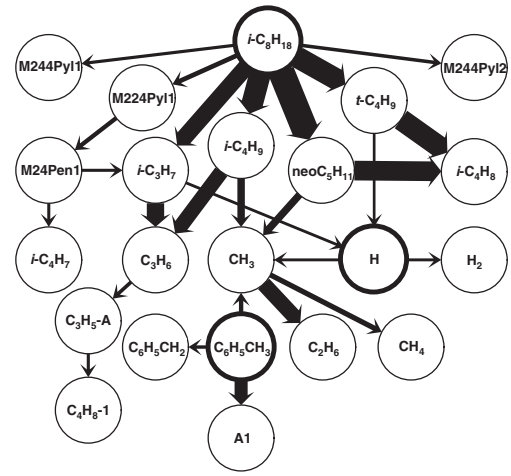

Fig. 7. Reaction path diagram from MEC2 [23] at the position of maximum C_6H_6 for $R_T = 0.8$ for *iso*-octane/toluene mixture flame in SF condition.

For A2 ($C_{10}H_8$), both mechanisms show monotonic trends, but the major reactions are different as $A1CH_2$ (C_7H_7) + $C_3H_3 \rightarrow$ A2 ($C_{10}H_8$) + H_2 (MEC1) and A1- (C_6H_5) + $C_4H_4 \rightarrow$ A2 ($C_{10}H_8$) + H (MEC2). For A3 ($C_{14}H_{10}$), the major reaction is $C_9H_7 + C_5H_5 \rightarrow$ A3 ($C_{14}H_{10}$) + 2H in MEC1, while A1- (C_6H_5) + $A1C_2H$ (C_8H_6) $\rightarrow$ A3 ($C_{14}H_{10}$) + H and P2- ($C_{12}H_9$) + $C_2H_2 \rightarrow$ A3 ($C_{14}H_{10}$) + H in MEC2. Although A1- and P2- increase with R_T in MEC2, $A1C_2H$ shows significantly pronounced synergistic trend, and C_2H_2 decreases with R_T, leading to the synergistic behavior of A3 in MEC2. For A4 production, the major reaction is the same in both mechanisms, A3- ($C_{14}H_9$) + $C_2H_2 \rightarrow$ A4 ($C_{16}H_{10}$) + H. In MEC2, A3- shows synergistic behavior. But in MEC1, it increases with R_T, resulting in the monotonic trend of A4 although C_2H_2 decreases.

3.2. Soot formation/oxidation flames

Figure 8 shows LII and LIF profiles for the SFO flames of *iso*-octane/toluene mixture. The maximum LII signal increases monotonically with R_T. The different aspects for the present SFO case as compared to SF case in terms of respective positions are: (1) the location of maximum LII signal which is much closer to the position of blue flame near 2.64 mm from the fuel nozzle due to the nature of soot formation and oxidation, and (2) the width of soot zone is somewhat narrower as compared to the SF flame cases. These points are expected since soot particles are migrating toward the flame.

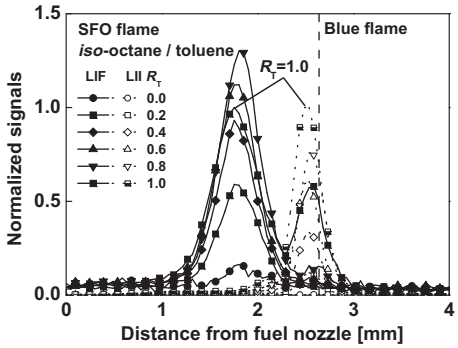

Fig. 8. Profiles of normalized LII and LIF signals for *iso*-octane/toluene mixture flames at several toluene ratios for SFO flames.

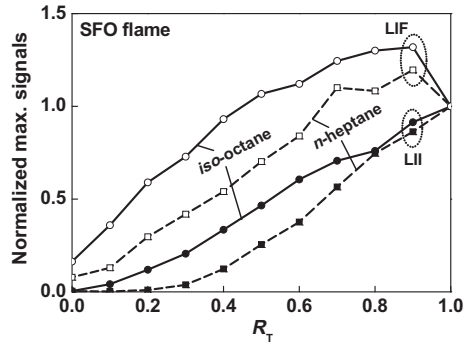

Fig. 9. Normalized maximum LII and LIF signals as a function of toluene ratio in *n*-heptane/toluene and *iso*-octane/toluene mixture flames in SFO condition.

The characteristics of LIF signals exhibit markedly different behavior as compared to the SF flames, where the PAH signals were monomodal. For the present SFO flames, however, the PAH profiles (400 nm) demonstrate bimodal nature with distinct double peaks, especially for large toluene ratios. This type of bimodal profile has been reported previously [29]. This can be explained based on the soot formation and oxidation nature of SFO flames. The LIF signal starts to increase near 1.2 mm from the nozzle. As the soot particles are formed, say near 2.0 mm as seen in the LII profiles, the LIF signal starts to decreases. Near 2.5 mm, where the LII signal starts to decrease, the LIF signal starts to increase again. This result is in accordance with the fact that soot particles are produced from PAHs and soot oxidation produces PAHs.

Note that the maximum in the first peak close to the fuel nozzle demonstrates a synergistic effect having the peak value near $R_T = 0.8$, which is higher than that for 1.0. While the second peak shows monotonic behavior with the toluene ratio. This can be understood based on the monotonic behavior of the LII signal with the toluene ratio, since the second peak is expected to be from soot oxidation. The cases of *n*-heptane/toluene mixtures demonstrate qualitatively similar behavior as the cases of *iso*-octane/toluene mixtures, although not shown.

The normalized maximum LII and the first peak LIF signals as a function of toluene ratio for *n*-heptane/toluene and *iso*-octane/toluene fuels for SFO flames are shown in Fig. 9. The maximum LII signal exhibits similar tolerance behavior as the cases for the SF flames in Fig. 4, in that for the *n*-heptane/toluene mixtures, the LII signals are minimal at small toluene ratios, while such behavior is not conspicuous for the *iso*-octane/toluene case. As compared to the cases of the SF flames in Fig. 4, the tolerance values are smaller for the SFO flames.

The maximum LIF signals for the SFO flames show similar behavior to the SF flames, demonstrating the synergistic non-monotonic behavior. The peak values in the LIF signals are shifted to higher toluene ratio as compared to the SF case. The results in Figs. 4 and 9 show that *iso*-octane has relatively higher tendency in producing soot and PAHs as compared to *n*-heptane, when mixed with toluene.

Figure 10 shows the numerical result for the normalized maximum mole fractions of aromatic rings in SFO flames for the *iso*-octane/toluene mixture by using MEC2. The benzene ring formation shows the synergistic behavior. The result with MEC1 exhibits similar behavior although not shown. When using MEC1, the synergistic effect is much weaker and the effect disappears for larger rings of A2, A3, and A4, which increase monotonically with R_T. With MEC2, A3 and A4 show the synergistic behavior. The peak mole fraction of A4 for *iso*-octane/toluene mixture is about 1.25 at $R_T \approx 0.8$. This value is similar to the experiment in Fig. 9, with the peak about 1.3 at $R_T \approx 0.9$. The numerical results also

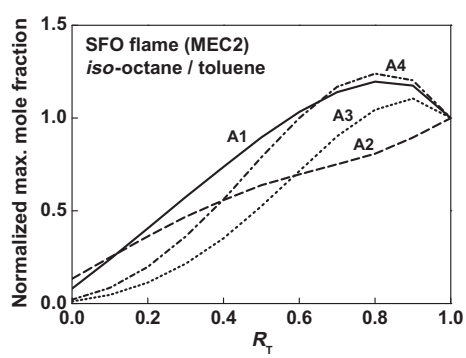

Fig. 10. Normalized maximum mole fractions A1, A2, A3 and A4 calculated by MEC2 [23] as a function of toluene ratio for *iso*-octane/toluene mixture flames in SFO condition.

indicate that the peak in A4 for the SFO flame appears at higher toluene ratio as compared to the SF cases.

When comparing with the cases of SF flames, the synergistic behavior was better predicted quantitatively for the present SFO flames. The reason may be attributed to the fact that MEC2 is validated for rich premixed flat flames and jet stirred reactors. In a counterflow configuration, it has been demonstrated [29] that soot models based on premixed flame can reasonably predict the characteristics of SFO flames because of the existence of oxidizing species. This could understand the appreciable over-prediction in the synergistic effect for the SF flames shown in Fig. 5. The present results imply that the kinetic mechanisms require further refinements to be applicable to both SF and SFO flames of toluene fuel mixed with *n*-heptane and *iso*-octane.

4. Concluding remarks

The experiment has been conducted for the three binary mixtures of the three surrogate fuels of gasoline of *n*-heptane, *iso*-octane, and toluene. For the present conditions of experiments for the SF and SFO flames, *n*-heptane, *iso*-octane, and their mixtures were found to be non-sooting. When toluene was mixed to either *n*-heptane or *iso*-octane, there existed a tolerance in the mixing of toluene below which the effect of toluene on soot formation was minimal.

The profiles of PAH LIF signals were different for the SF and SFO flames by having mono-modal and bimodal nature, respectively. This is due to the characteristics of soot formation without oxidation and soot formation followed by soot oxidation for the SF and SFO flames, respectively. For both types of flames, the PAH LIF signals exhibited synergistic effect in term of toluene ratio. Numerical simulation could predict the synergistic effect for benzene ring formation as a soot precursor. The H radical, which is important in benzene ring formation from toluene, decreases when the toluene ratio increases. This effect leads to the decrease in benzene ring at large toluene ratios. For larger aromatic rings, the numerical results with MEC2 could predict the synergistic effect for A4, while the effect is highly over-predicted for SF flames. The present result may provide a useful data for the soot modeling for the gasoline surrogate fuels and fuel formulation in the future.

Acknowledgment

This work was supported by Saudi Aramco through KAUST CCRC.

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